MIL-S-63433C(AR)
24 April 1987
SUPERSEDING
MIL-S-63433B(AR)
10 May 1985

MILITARY SPECIFICATION

SIMULATOR, FLASH, ARTILLERY, M21, PERFORMANCE SPECIFICATION FOR

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for design and establishes a uniform method of testing one type of Artillery Flash Simulator, M21. The purpose of the testing program is to determine performance, safety, soundness of design and resistance to environments encountered during storage, handling, and service use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation - Packaging, Methods Of.
MIL-A-2550	- Ammunition and Special Weapons; General Specification For.
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1370
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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	- Quality Assurance Terms and Definitions.
MIL-STD-331	- Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1234	- Pyrotechnics, Sampling, Inspection and Testing.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9342970	- Packing & Marking for Box, Paperboard, for Simulator Flash, Artillery, M21.
9342972	- Packing & Marking for Box, Packing, for Simulator, Flash, Artillery, M21.

TEST PROCEDURES

TOP 1-2-511	- Electromagnetic Radiation Effects and/or Hazards Test.
TOP 4-2-813	- Static Fragmentation Test of High Explosives.
TB 700-2	- DoD Explosive Hazard Classification Procedures.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-B-117 - Salt Spray (Fog) Testing.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Assembly. The simulator assembly shall correctly interface with the firing device manufactured by Hoffmann-Werke designated as Simulator, Tank, Gun Fire Device 17-61 and shall comply with all requirements specified in applicable specifications and standards.

3.2 Materials.

3.2.1 Nonmetals. Nonmetal material such as sealants and adhesives shall be a type that will not deteriorate with age or change properties to the extent that the simulators fail to conform to the functional requirements specified under 3.4 of this specification at any time during the warranty period, or five years whichever is longer when packaged as specified in 5.4.

3.2.2 Metals. Metals shall be corrosion-resistant or shall be suitably treated to resist corrosion due to salt spray when tested as specified in 4.5.7.

3.2.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.2.4 Compositions. All compositions (including but not limited to propellants, ignition compositions, flash compositions, transfer compositions, and first fire compositions) shall be stable and shall be compatible with the other compositions and materials of construction that they come in contact with. All energetic compositions shall be subjected to the following tests.

3.2.4.1 Impact. The composition shall be subjected to the impact test as specified in 40.1 of Appendix A to this specification. The 50% point shall not be less than 20 centimeters.

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3.2.4.2 Friction pendulum. The composition shall be subjected to the friction pendulum test as specified in 40.2 of Appendix A to this specification. The energetic material may burn, but may not detonate.

3.2.4.3 Electrostatic discharge. The composition shall be subjected to the electrostatic discharge test as specified in 40.3 of Appendix A to this specification. There shall be 20 out of 20 no-fires at 0.25 joules.

3.2.4.4 Explosive temperature. The composition shall be subjected to the explosive temperature test as specified in 40.4 of Appendix A to this specification. The explosive temperature shall be 200 degrees centigrade (^{o}C) minimum (min.).

3.2.4.5 Vacuum thermal compatability. The composition shall be subjected to the vacuum thermal compatability test as specified in 40.5 of Appendix A to this specification. There shall be not more than 2 milliliters of gas evolved per 5 grams of energetic material.

3.2.4.6 Auto-ignition temperature. The composition shall be subjected to the auto-ignition temperature test as specified in 40.6 of Appendix A to this specification. The auto-ignition temperature shall be 200°C min.

3.2.4.7 Differential thermal analysis. The composition shall be subjected to differential thermal analysis as specified in 40.7 of Appendix A to this specification.

3.2.4.8 Rotary friction. The composition shall be subjected to the rotary friction test as specified in 40.8 of Appendix A to this specification. EQ shall be greater than 100 foot pounds per second, where EQ=WT (0.000544).

3.2.5 Safety. This specification contains requirements which may necessitate the handling of hazardous materials and result in the manufacture of a hazardous item. It is emphasized that all applicable safety rules, procedures, and regulatory requirements be strictly followed during the handling, processing, and testing of the completed item and related hazardous materials. All required safety information shall be submitted to the Contracting Officer for approval.

3.2.5.1 Department of transportation (DOT). DOT authorization and classification to ship shall be obtained by the supplier. A copy of the DOT authorization shall be furnished to the Contracting Officer prior to delivery of any assemblies, items, or hazardous materials. The simulator, in its shipping and storage packaging, shall meet the requirements of the following classifications:

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DoD Hazard Classification - 1.3
DoD Compatability Group - G
DoT Hazard Classification - Class C Explosives
DoT Marking - SPECIAL FIREWORKS - HANDLE CAREFULLY - KEEP FIRE AWAY

When tested in accordance with 5.3.f "Stack Test" and 5.3.g "External Fire Stack Test" of TB 700-2.

3.2.5.2 DOT authorization/hazard classification. Certification from the supplier shall be required to assure no significant change in the item, materials, packaging, etc. which would affect the DOT authorization/hazard classification.

3.2.5.3 Hazardous materials (energetic, toxic, carcinogenic). Hazardous materials/components will be identified as they are selected, and submitted to the Contracting Officer for approval. Only energetic materials acceptable for DOD use will be utilized.

3.2.5.4 Explosive ordnance disposal (EOD). EOD render safe procedures and applicable render safe procedure tools will be formulated as early as possible and will be kept current with design changes of the simulator as they are instituted. These procedures and tools shall be made available to the Government for approval as early as possible to insure applicable tools, equipment and publications will be available to army EOD units thirty (30) days before the deployment date of the simulator.

3.2.5.5 Demilitarization. Demilitarization/disposal procedures will be formulated and kept current with design changes as they are instituted. These procedures shall be made available to the Government.

3.2.5.6 Static and other induced electric hazards. The ignition system should be designed and maintained so that static charges and induced electric energies will not produce a hazardous device.

3.2.5.6.1 Electromagnetic radiation susceptibility. The simulator shall not function when tested as specified in 4.5.4.4.

3.2.5.6.2 Electrostatic discharge (ESD). The simulator shall not function from human-borne or aircraft-borne ESD when tested as specified in 40.3 of Appendix A.

3.3 Design and construction. The components of the simulator shall be designed and constructed to conform to the performance requirements specified herein.

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3.3.1 Assembly of components. The components of the simulator shall consist of at least a container, flash charge, igniter, a cable assembly and some method of protecting the item from inadvertant or accidental initiation.

3.3.2 Application of sealants and adhesives. Any sealants or adhesives used in the construction of the simulator shall be applied in such a manner that the performance of the simulator in 3.4 after prolonged storage shall not be degraded by aging or any other changes of the sealant or adhesive, e.g., a sealant used to seal the interface of a cap and container shall be applied in such a manner that excessive sealant will not enter the interior of the simulator and cover or encapsulate the chemical components in such a way as to change the functional characteristics of the simulator under prolonged storage.

3.3.3 Plastic parts. The use of plastic parts shall be subject to the approval of the Contracting Officer for the specific application involved.

3.3.4 Performance characteristics. The simulator shall not damage the firing device or interfere with repeated use.

3.3.5 Moisture content.

3.3.5.1 Charge composition. The moisture content of the charge composition at the loading station at the time of loading shall not exceed 0.10 percent when tested as specified in 4.5.3.1.

3.3.5.2 Paper and paper products. The moisture content of any paper and paper product components of the simulator assembly at the loading station at the time of loading shall not exceed 6.0 percent when tested as specified in 4.5.3.2.

3.3.6 Performance reliability. The simulator must meet all performance requirements at a reliability of not less than 95 percent at a confidence level of 90 percent over a temperature range of 125 $\pm$ 5°F to minus 25 $\pm$ 5°F as defined in 4.3.3 and 4.4.2.

3.4 Performance. The simulator shall satisfy all requirements specified herein after subjection to the applicable environmental tests. Performance shall be measured and values recorded in accordance with 6.4 for the ballistic operation of the simulator.

3.4.1 Functioning. The assembly shall function and comply with the requirements of 4.5.4 after conditioning at the specified temperatures.

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3.4.1.1 Sound characteristics. The sound level intensity of the simulator shall be a minimum (min.) of 124 decibels (db) to a maximum (max.) of 147 decibels (db) when measured at a distance of 96 meters min. from the item under test when tested per 4.5.4.1.

3.4.1.2 Functioning time. The time from application of twenty-four (24) plus or minus (±) two (2) Volts Direct Current (VDC), with minimum ignition energy of 3.0 milliwatt seconds/ohm, to initiation of the assembly shall not be greater than 0.20 seconds (sec.) when tested as specified in 4.5.4.2.

3.4.1.3 Flash characteristics. The flash intensity of the simulator shall be 4×10^6 candlepower minimum (min.) when tested as specified in 4.5.4.3. Residual smoke shall be white in appearance.

3.4.2 Chamber withdrawal. The simulator shall meet a pull test requirement of 2.8 to 8 pounds (lbs.) force at a temperature of 70 ± 5°F when tested as specified in 4.5.2.

3.4.3 Bridge-wire resistance. The igniter bridge-wire resistance shall be 1.3 ± .2 ohms when tested as specified in 4.5.9.

3.4.4 Functional timing sequencing of the electric igniter. The continuity of the electric igniter shall be 10 milliseconds (ms) minimum at 12 VDC (nominal) at 0.5 amperes (amp). It must produce an open circuit (if only for an instant) within 25 ms of receiving 12 VDC at 1 ampere when tested as specified in 4.5.10.

3.4.5 Surface danger zone. When fired the simulator shall produce a hazardous fragment distance of no greater than twenty-five (25) meters to either side or fifty (50) meters forward of the firing device when tested as specified in 4.5.12. A fragment will be considered hazardous if it contains 58 ft. lbs/sec kinetic energy at the specified distance.

3.5 Environmental conditions.

3.5.1 Vibration. The simulators, when packed, shall not fire when subjected to vibration as specified in 4.5.8.1. After subjection to such vibration (and additional testing as indicated for first article tests), the simulators shall show no signs of damage, shall be safe to handle and fire, and shall meet the performance requirements of 3.4.1.

3.5.2 Shock. The simulators, when packed, shall not fire when subjected to shock testing as specified in 4.5.8.4. After subjection to such shock testing (and additional testing as indicated for first article tests or quality conformance inspection), the simulators shall show no signs of damage, shall be safe to handle and fire, and shall meet the performance requirements of 3.4.1.

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3.5.3 Seven-foot drop. The simulators, when packed, shall not fire when subjected to drops from a height of seven feet as specified in 4.5.8.6. After being subjected to such drops (and additional testing as indicated for first article tests), the simulators shall show no signs of damage, shall be safe to handle and fire, and shall meet the performance requirements of 3.4.1.

3.5.4 Loose cargo. The simulators, when packed, shall not fire when subjected to loose cargo testing as specified in 4.5.8.8. After being subjected to the loose cargo test (and additional testing as indicated for first article tests, or quality conformance inspection), the simulators shall show no signs of damage, shall be safe to handle and fire, and shall meet the performance requirements of 3.4.1.

3.5.5 Ten foot drop. The unpackaged simulators shall not fire when dropped from a height of ten (10) feet as specified in 4.5.8.5. After being subjected to such a drop, the simulators if not damaged to the extent that they cannot be loaded into the firing device, must be safe to fire. If after such a drop, the simulators are damaged to the extent that they cannot be loaded into the firing device, they shall be safe to handle and dispose of.

3.5.6 Forty-foot drop. The simulators, when packed, shall not fire when dropped from a height of forty feet as specified in 4.5.8.7 and shall be safe to handle and be disposed of.

3.5.7 Low temperature. The simulators shall be capable of withstanding low temperature conditions as specified in 4.5.8.2 and shall meet the performance requirements of 3.4.1.

3.5.8 High temperature. The simulators shall be capable of withstanding high temperature conditions as specified in 4.5.8.3 and shall meet the performance requirements of 3.4.1.

3.5.9 Leakage test. The simulators, when packaged, shall be capable of withstanding exposure to the leakage test as specified in 4.5.6 and shall meet the performance requirements of 3.4.1.

3.5.10 Water immersion test. The simulator shall be capable of withstanding exposure to water immersion as specified in 4.5.8.11 and shall meet the performance requirements of 3.4.1.

3.5.11 Jumble. The simulator shall comply with the following requirements when tested as specified in 4.5.5.

a. The simulator shall not function during test.

b. There shall be no powder leakage.

c. The simulator components shall not separate during test.

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3.5.12 Temperature-humidity. The packed simulator shall withstand exposure to temperature-humidity cycle as specified in 4.5.8.9 and shall meet the performance requirements of 3.4.1.

3.5.13 Temperature cycling. The simulator shall be subjected to the temperature cycling as specified in 4.5.8.10 and shall meet the performance requirements of 3.4.1.

3.6 Five foot drop. The unpackaged simulator shall not fire when dropped from a height of five feet as specified in 4.5.8.12. After being subjected to such a drop, the simulator shall be safe to handle and fire, and shall meet the applicable requirements of 3.4.1.

3.7 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, crazes, porosity, warpage, burn marks, blisters, excess flash, scratches, dirt, grease and other defects and foreign matter which could affect appearance or serviceability. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined. All packing components shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any or all of the inspections set forth in the specification, where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Materials. Inspection and testing of component parts and assemblies shall be made to determine compliance with 3.2 and 3.3. If material defects, incompatability, or inferior quality is evident, and the Government deems material analysis necessary, the contractor may be requested to submit samples or specimens to the cognizant design agency for analysis and approval.

4.2 Classification of tests. The following types of inspection shall be performed on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of Table I.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in this specification.

4.3.3 Rejection.

a. If an assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon failure of any assembly, component, or test specimen in the sample to comply with any of the stated requirements.

b. Each of the environmental test samples shall pass the accept/reject criteria for each specified environment.

c. The sum of all samples tested (12 functioning tests for a total of 918 samples) shall pass an accept/reject criteria of 918-30-31.

d. A simulator shall be classified as a functional failure if it fails any of the requirements of 3.4.1.

e. Any simulator, upon functioning, that damages the firing device making any tube unusable shall reject the First Article.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-S-63433C(AR)

PARAGRAPH	TITLE: Simulator, Flash, Artillery, M21		SHEET 1 OF 3		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
			Ac/Re		
	Bridge-wire resistance	20	0-1	3.4.3	4.5.9
	Functional timing sequencing test	20	0-1	3.4.4	4.5.10
	Chamber withdrawal test	20	0-1	3.4.2	4.5.2
	Moisture content				
	Charge composition	25	0-1	3.3.5.1	4.5.3.1
	Paper or paper products	25	0-1	3.3.5.2	4.5.3.2
	Hazard Classification				
	Stack Test	15 boxes		3.2.5.1	
	External Fire Stack Test	5 boxes		3.2.5.1	
	Electromagnetic radiation	1 box	0-1	3.2.5.6.1	4.5.4.4
	Electrostatic Discharge	1 box	0-1	3.2.5.6.2	40.3 Appendix A
	Surface Danger Zone	1		3.4.5	4.5.12
	Ten Foot Drop	30	0-1	3.5.5	4.5.8.5

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-S-63433C(AR)

PARAGRAPH	TITLE: Simulator, Flash, Artillery, M21		SHEET 2 OF 3		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
			Ac/Re		
	Non-sequential test				
	Temperature cycling			3.5.13	4.5.8.10/4.5.4
	Cold, Ambient	30	1-2		
	Cold, Ambient, Hot	30	1-2		
	Cold, Hot	30	1-2		
	Cold, Hot, Ambient	30	1-2		
	Hot, Ambient	30	1-2		
	Hot, Cold	30	1-2		
	Jumble - Ambient	30	0-1	3.5.11	4.5.5
	Forty foot drop			3.5.6	4.5.8.7
	Cold	162 (1 box)	0-1		
	Hot	162 (1 box)	0-1		
	Temperature-humidity	162 (1 box)		3.5.12	4.5.8.9
	Function	30	1-2		4.5.4
	Low Temperature	30	1-2	3.5.7	4.5.8.2/4.5.4
	High Temperature	30	1-2	3.5.8	4.5.8.3/4.5.4
	Heat seal of barrier bag	18 (1 box)		3.5.4	4.5.11
	Sequential Test				
	Leakage test (prior to shipment)	648 (4 boxes)		3.5.9	4.5.6

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-S-63433C(AR)

PARAGRAPH	TITLE: Simulator, Flash, Artillery, M21		SHEET 3 OF 3		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
			Ac/Re		
	Vibration			3.5.1	4.5.8.1
	Cold	324 (2 boxes)	0-1		
	Hot	324 (2 boxes)	0-1		
	Shock			3.5.2	4.5.8.4
	Cold	324 (2 boxes)	0-1		
	Hot	324 (2 boxes)	0-1		
	Seven foot drop			3.5.3	4.5.8.6
	Cold	324 (2 boxes)	0-1		
	Hot	324 (2 boxes)	0-1		
	Loose cargo			3.5.4	4.5.8.8
	Cold	324 (2 boxes)	0-1		
	Hot	324 (2 boxes)	0-1		
	Leakage test	648 (4 boxes)	0-1	3.5.9	4.5.6
	Five foot drop			3.6	4.5.8.12
	Cold (Note 1)	30	1-2		4.5.4
	Hot (Note 1)	30	1-2		4.5.4
	Water immersion test	648		3.5.10	4.5.8.11
	Functioning-Ambient	588	20-21	3.4.1	4.5.4

NOTES:
(1) Fifteen (15) of these items shall be taken from each of the respective boxes. These items shall be subjected to water immersion prior to functioning.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. See MIL-A-48078.

4.4.2 Examination. The examinations/tests listed herein shall be performed on inspection lots as defined in 4.4.1 of this specification.

a. Sampling plans. Unless otherwise specified in the classification of defects or test tables, sampling plans and procedures shall be in accordance with MIL-STD-105, Level II.

b. Each of the environmental test samples shall pass the accept/reject criteria for each specified environment.

c. The sum of all samples tested (four functioning tests for a total of 180) shall pass an accept/reject criteria of 180-4-5.

d. A simulator shall be classified as a functional failure if it fails any of the requirements of 3.4.1.

e. Any simulator, upon functioning, that damages the firing device making any tube unuseable shall reject the lot.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63433C(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Simulator, Flash, Artillery, M21	1 OF 2	NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
			Ac/Re	Simulator	
	Physical characteristics and dimensions (Major or minor defects as applicable)	81	0-1	3.1	4.5.1
	Bridgewire resistance test (Major defect)	81	0-1	3.4.3	4.5.9
	Functional timing sequencing test (Major defect)	81	0-1	3.4.4	4.5.10
	Chamber withdrawal test (Major defect)	81	0-1	3.4.2	4.5.2
	Moisture content (Major defect)				
	Charge composition	25	0-1	3.3.5.1	4.5.3.1
	Paper products	25	0-1	3.3.5.2	4.5.3.2
	Water immersion test - Note 4 (Major defect)	150	0-1	3.5.10	4.5.8.11
	Non-sequential test - Note 1				
	High temperature (Major defect)	50	1-2	3.5.8	4.5.8.3 4.5.4
	Low temperature (Major defect)	50	1-2	3.5.7	4.5.8.2 4.5.4

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63433C(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.1	Simulator, Flash, Artillery, M21				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
			Ac/Re		
	Ambient (Major defect)	50	1-2	3.4.1	4.5.4
	Total (Major defect)	150	3-4		
	Jumble (Critical defect)	25	0-1	3.5.11	4.5.5
	Heat seal of barrier bag - subpacks (Major defect)	18	0-1	3.5.4	4.5.11
	Sequential test - (Ambient) Note 2				
	Leakage test (Major defect)	81	0-1	3.5.9	4.5.6
	Shock/6 drops - Note 3 (Major defect)	81	0-1	3.5.2	4.5.8.4
	Loose cargo (Major defect)	81	0-1	3.5.4	4.5.8.8
	Leakage test (Major defect)	81	0-1	3.5.9	4.5.6
	Five foot drop (Critical defect)	30	0-1	3.6	4.5.8.12
	Functioning - Ambient (Major defect)	30	1-2	3.4.1	4.5.4

NOTES: (1) Low minus 25 ± 5°F, Ambient 70 ± 5°F, High +125 ± 5°F. To be conditioned for six (6) hours min. and fired within five (5) minutes of removal for temperature chamber.

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4.4.3 Testing. Testing is described in the first article and quality conformance inspection tables.

4.4.3.1 Moisture content (see Table II) - Major Defects.

TABLE II - Moisture Content

Charge composition	(see 3.3.5.1)
Paper and paper products	(see 3.3.5.2)

The Contractor shall provide adequate controls to insure that the materials comply with the requirements. For verification, the contractor shall select and subject to test, one sample of each material from each eight (8) hours production. If a composite sample exceeds the requirement, that quantity or sub-lot material represented by the sample shall not be used in loading or packing. The remaining unloaded and unpacked material shall not be used in production, and the loaded and packed simulators shall be rejected. Test shall be performed as specified in 4.5.3.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with Table I and 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See 6.3 and 6.4 herein.

4.5 Methods of inspection.

4.5.1 Examination of product. Examination of the simulators shall be made to detect visible defects and to determine that the items meet the requirements of this specification with respect to finish and marking, and that they conform to the applicable drawing.

4.5.2 Chamber withdrawal test. The simulators, after being installed in the approved firing device, shall be subjected to a pull force test. The approved firing device tube shall be clean and dry. Slowly apply a pulling force on the simulator using an approved method in accordance with 4.4.4. The pull force required to extract the simulator from the tube shall be in compliance with the applicable requirement as specified in 3.4.2. (Non-destructive test).

4.5.3 Moisture content.

4.5.3.1 Charge composition. The method of determining the percentage of moisture in the flash charge shall be in accordance with Test Method 102.2.1 of MIL-STD-1234 except that the sample shall be heated until a constant weight is obtained in lieu of four hours. Observation shall be made for compliance with the applicable requirement as specified in 3.3.5.1.

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4.5.3.2 Paper or paper products. The moisture content shall be determined in accordance with Test Method 102.1.1 of MIL-STD-1234 except that a ten (10) gram sample shall be used and the drying time shall be for two (2) hours. Observation shall be made for compliance with the applicable requirements as specified in 3.3.5.2.

4.5.4 Functioning. The simulator shall be conditioned for 6 hours min. at the specified temperatures and fired within five (5) minutes after removal from the conditioning chamber. The simulator shall be mounted horizontally in an approved firing device and fired by means of an approved power source in accordance with 3.4.1.2. (Destructive test)

4.5.4.1 Sound characteristics. Measurement shall be made from a side position to the longitudinal axis of the simulator which is to be suspended horizontally 4.9 $\pm$ .2 meters off the ground. The detector shall be positioned 1.5 $\pm$ 0.1 meters off the ground for measurement purposes. Observation shall be made for compliance with the applicable requirement of 3.4.1.

4.5.4.2 Functioning time. The time from the application of the 24 VDC to the item functioning shall be measured with approved metering equipment in accordance with 4.4.4. The simulator shall be observed for compliance with the applicable requirement of 3.4.1.2.

4.5.4.3 Flash characteristics. The flash intensity of the simulator shall be recorded using an approved method in accordance with 4.4.4. The simulator and detector positioning shall be the same as 4.5.4.1. Observation will be made for conformance with the applicable requirements of 3.4.1.

4.5.4.4 Electromagnetic radiation susceptibility. The simulator, bare, packaged, and in its firing device shall not function when tested in accordance with TOP 1-2-511, Appendix A.

4.5.5 Jumble. The simulator shall be tested in accordance with procedures specified in MIL-STD-331, Test 102, except that the duration of the test shall be a minimum of 20 minutes. The simulator and equipment shall be observed and inspected for compliance with the applicable requirements. (Destructive Test).

4.5.6 Leakage test. The test shall be performed in accordance with the procedure specified in 4.4.3.1 and 4.4.3.2 or 4.4.3.6 of MIL-P-116. After the test, the seal shall be examined for compliance with the applicable requirements.

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4.5.7 Salt spray test. The specified parts shall be subjected to the salt spray test in accordance with ASTM-B-117 using a 5% atomized sodium chloride solution. Exposure time shall be not less than two (2) hours. Examination shall be made for any evidence of corrosion or salt deposits indicative of improper coating. Any evidence of improper coating shall cause the part to be classed defective. (Destructive test)

4.5.8 Environmental tests. Environmental testing shall be conducted to determine compliance with the requirements of 3.5. The packed configuration used in the subsequent tests shall be in accordance with 5.5.

4.5.8.1 Vibration. This test shall be conducted in accordance with Method 514.3 MIL-STD-810, Test Procedure I, Test Condition I-3.2.1, Figures 514.3-1, 514.3-2, 514.3-3 (Vertical, Transverse and Longitudinal axis) for 120 minutes (equal to 2000 miles) in each axis. Observation shall be made for compliance with the applicable requirement.

4.5.8.2 Low temperature. This test shall be conducted in accordance with Method 502.2, Procedure I of MIL-STD-810, except that the simulator shall be exposed to a temperature of minus 25 $\pm$ 5°F until thermal stabilization is reached and maintained for a period of 6 hours minimum. The test specimen is then removed from the conditioning cabinet, set up, and fired within a period of time not to exceed 5 minutes. Observation shall be made for compliance with the applicable requirement.

4.5.8.3 High temperature. This test shall be conducted in accordance with Method 501.2, Procedure I of MIL-STD-810, except that the simulators shall be exposed to a temperature of 125 $\pm$ 5° until thermal stabilization is reached and maintained for a period of 6 hours min. The test specimen is then removed from the conditioning cabinet, set up, and fired within a period of time not to exceed 5 minutes. Observation shall be made for compliance with the applicable requirement.

4.5.8.4 Shock. This test shall be conducted in accordance with Method 516.3, Procedure IV of MIL-STD-810, Table 516.3-II, except that drop height shall be 36 inches. Use column with largest dimension - 91 centimeters (cm) and over and weight under 45.4 kilograms (kg) (100 lbs). The packed simulators shall be observed for compliance with the applicable requirement.

4.5.8.5 Ten foot drop test. The simulator shall be dropped from a height of 10 feet onto a three inch thick steel plate embedded in concrete. During First Article Testing, thirty items will be tested. Six items will be dropped for each of the five orientation listed below. Observation shall be made for compliance with the applicable requirements.

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a. Bottom down.
b. Top down.
c. Side down.
d. Bottom down at 45° angle.
e. Top down at 45° angle.

4.5.8.6 Seven foot drop test. The packed simulators shall be dropped from a height of 7 feet (measured to the bottom of the package) onto a 3-inch thick steel plate embedded in concrete in the orientations below:

a. One drop on side and one drop on end; at high temperature.
b. One drop on top and one on bottom edge at 45° angle; at high temperature.
c. One drop on bottom and one drop on top edge at 45° angle; at low temperature.
d. One drop on side (opposite (a) side) and one drop on one end (opposite (a) end); at low temperature.

The packed simulators shall be observed for compliance with the applicable requirement.

4.5.8.7 Forty-foot drop test. This test shall be conducted in accordance with Test 103 of MIL-STD-331. The packed simulators shall be dropped in the following positions:

a. One pack dropped with bottom down at low temperature.
b. One pack dropped with end down at high temperature.

The simulator shall be observed for compliance with the applicable requirement. (Destructive Test)

4.5.8.8 Loose cargo. This test shall be conducted in accordance with Method 514 of MIL-STD-810, Test Procedure II, Test Conditions I-3.2.3. Two boxes shall be positioned on end, one at high and one at low temperature, and two boxes shall be positioned on their base, one at high and one at low temperature. The simulators shall be observed for compliance with the applicable requirement.

4.5.8.9 Temperature humidity. The packed simulators shall be subjected to a 14 day temperature humidity test according to Test 105.1 of MIL-STD-331. Observation shall be made for compliance with the applicable requirement.

4.5.8.10 Temperature cycling. The simulator shall be conditioned at the applicable temperature and cycled as specified in MIL-STD-810, Method 503.2, Table 503.2-1, hot dry ambient (70 ± 5°F) temperature. Observation shall be made for compliance with the applicable requirements.

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4.5.8.11 Water immersion test. The unpackaged simulator, without cap, shall be submerged in 12 inches minimum of water at ambient (70 ± 5°F) temperature for a minimum of one hour. The simulator shall then be removed from the water, placed in the firing device, and functioned. The simulator shall comply with the applicable requirements. (Destructive Test)

4.5.8.12 Five foot drop test. The simulators shall be fired after a drop from five feet onto a three inch thick steel plate embedded in concrete. Thirty items will be tested at high temperature and thirty items tested at low temperature. Six items in each temperature phase will be dropped for each of the five orientations listed below. Observation shall be made for compliance with the applicable requirements.

a. Bottom down.
b. Top down.
c. Side down.
d. Bottom down at 45° angle.
d. Top down at 45° angle.

4.5.9 Bridge wire resistance test. The igniter shall be tested for the required bridge-wire resistance with approved test equipment in accordance with 4.4.4. Observation shall be made for compliance with the applicable requirement. (Non-Destructive Test)

4.5.10 Functional timing sequencing test. A continuity and open circuit test shall be performed on the electric igniter using an approved method and equipment in accordance with 4.4.4. Observation shall be made for compliance with applicable requirement. (Destructive Test)

4.5.11 Heat seal of barrier bag. (See Dwg. 9342970) The test shall be performed in accordance with the procedure specified in MIL-P-116 paragraph 4.4.4.3. After the test, the seal shall be examined for compliance with the applicable requirements (Destructive Test).

4.5.12 Fragmentation test. The simulators shall be subjected to the fragmentation test in accordance with TOP 4-2-813 except that item shall be fired from firing device and arena shall consist of a 200° arc extending 10° past perpendicular of firing device on both sides. Observation shall be made for compliance with the applicable requirements.

5. PACKAGING

5.1 Cleaning. The simulator shall be cleaned in accordance with Process C-1 of MIL-P-116, as required.

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5.2 Drying. The simulator shall be dried in accordance with MIL-P-116, as required.

5.3 Preservation. Not required.

5.4 Packaging and marking level. Packaging and marking shall be as specified on drawing 9342970. If the length of the simulator changes, additional fillers must be added to insure a tight fit or the width of the fiberboard box and length of the wood box changed accordingly.

5.5 Packing and marking level. Packing and marking shall be as specified on drawing 9342972.

6. NOTES

6.1 Intended use. The simulator is a device utilizing energy produced by explosives, propellants, or pyrotechnics or a combination thereof to produce a flash, bang and smoke signature. This is a training device used with the Hoffmann Device on several armored vehicles.

6.2 Ordering data. See contract.

6.3 Submission of inspection equipment for design approval. See MIL-A-48078. The contractor shall submit inspection equipment designs as required to Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-I (D), Picatinny Arsenal, NJ 07806-5000.

6.4 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.
3.4	Quality Inspection Test, Demonstration and Evaluation Report	DI-R-1724
4.4.4	Quality Engineering Acceptance Inspection Equipment Descriptive Documentation	DI-R-10054

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(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L. Contractors in connection with specific acquisition functions should obtain copies from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4.1 Warranted ammunition. When the Quality Assurance Contract/Section E contains requirement for a performance warranty, the ammunition data card, DD1650, shall include the following warranty information:

> Warranted ammunition: The performance of this lot of ammunition is under warranty until _____*_____.
>
> * The warranty expiration date for each lot shall be inserted by the contractor.

6.5 Contractor's design requirements and specifications.

6.5.1 Inert and explosive materials. All inert and explosive materials used in the simulator shall be referenced to approved Government specifications if such exist. Proprietary materials should not be used unless specifically approved in writing by the contracting officer. At the direction of the contracting officer, the contractor should submit simulator drawings, in sufficient detail as to parts, special processes, and techniques to permit the preparation of documentation in accordance with 6.5.2.

6.5.2 Drawings. Drawings should be in accordance with DOD-D-1000.

6.6 Noise protection. Noise protection is advisable to prevent ear damage whenever safe distance of 96 meters is not adhered to.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Subject term (key word) listing.
Simulator, M21
Flash
Sound
Moisture Content
Hazardous
Jumble

Custodian:
Army-AR

Preparing Activity:
Army-AR

(Project 1370-A267)

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SIMULATOR, FLASH, ARTILLERY, M21
PERFORMANCE SPECIFICATION FOR

10 SCOPE:

10.1 Scope. This appendix details the quality assurance test methods required for testing one type of Artillery Flash Simulator, M21. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

20 APPLICABLE DOCUMENTS.

20.1 Government documents.

20.1.1 Specifications and standards.

SPECIFICATIONS

MILITARY

MIL-STD-1234	- Pyrotechnics, Sampling, Inspection and Testing
MIL-STD-1751	- Safety and Performance Test for Qualification of Explosives

20.2 Other publications.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-E-474	- Evaluation of Temperature Scale for Differential Thermal Analysis
ASTM-D-710	- Low Carbon Steels Specification for

30. REQUIREMENTS. This section is not applicable to this appendix.

40. TEST METHODS.

40.1 Impact test (ERL Type 12 tools).

40.1.1 Apparatus. The apparatus shall include the following:

a. ERL machine (Bruceton Machine)

(1) Striker pin, 1.250 inch diameter by 3.500 inches long, made of tool steel, hardened to 60-63 Rockwell "C" scale. Flat surface next to explosive is ground to a finish of 16 microinches.

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(2) Flint sandpaper, 5/0 grade.

(3) Anvil, 1.250 inch diameter by 11.250 inches long, made of tool steel, hardened to 60 Rockwell "C" scale. All surfaces ground and polished, mounted in a tool holder assembly which is bolted to the machine base.

b. Microphone, Astatic Model JT-30C, ceramic type.

c. Thyratron tube, Model 2050.

d. Burlington Model 431 millivolt meter.

e. Weights, 2, 2.5 and 5 kilograms (weight commonly used is 2.5 kilograms), made of hardened steel. An electromagnet retains the drop weight until released by the operator.

40.1.2 Procedure.

40.1.2.1 Preparatiopn of apparatus.

a. Figure 1 shows the assembly of the test elements to be used, which consists of a free falling weight, tooling to hold the energetic sample, and a supporting frame.

b. The microphone is mounted in the horizontal plane of the anvil face at a distance of 34 inches from the center of the anvil.

c. Calibrate the millivolt meter across a 100 ohm resistor by adjusting the setting to 50 millivolts.

d. Adjust the amplifier gain to read 25 millivolts initially. Determine final adjustment by means of two test switches which make the thyratron tube alternately conductive and nonconductive. When proper gain setting has been achieved, as demonstrated by 10 or more consecutive tests.

e. Calibrate the apparatus at least once each week for proper elimination of background noise. Adjust the instrumentation as described in paragraph d above. Release the drop weight from maximum height to impact on the test anvil. Under these conditions the neon lamp must not glow.

40.1.2.2 Preparation of sample. The test energetics are solid, granular materials which are either pure compounds or mixtures.

a. For materials which are normally cast-loaded into a weapon, prepare by casting as a thin sheet (weight from 3 to 10 grams depending on material availability and number of determinations to be made).

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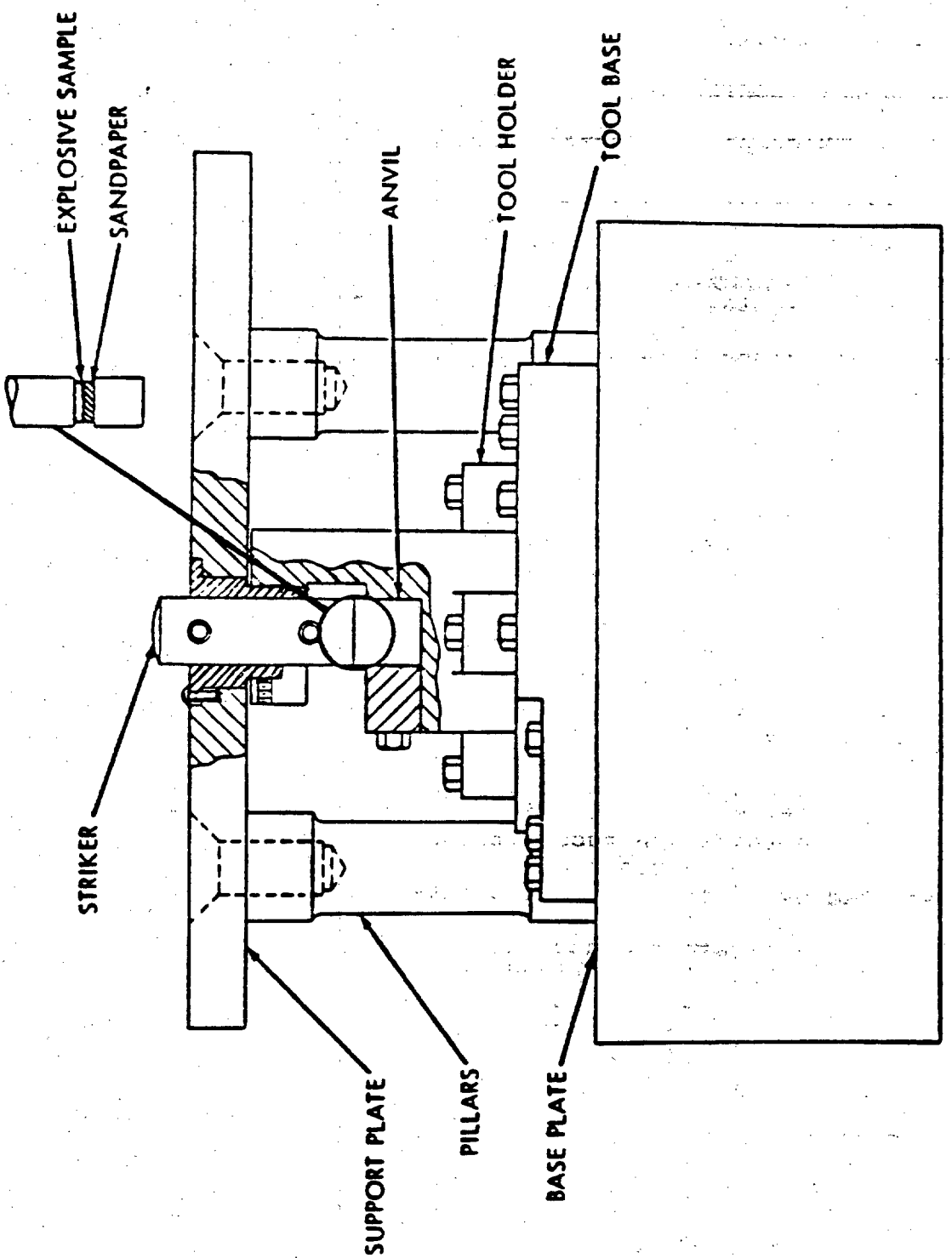


FIG. 1 IMPACT TEST ASSEMBLY

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(1) Grind the cast sheet gently by hand in a wooden mortar and screen the material through a set of No. 16, 30 and 50 U.S. Standard Sieves. Carefully blend equal wieghts of material retained on the No. 30 and No. 50 sieves on a Fisher-Kendall mixer (simultaneous) tumbling and stirring action) to furnish the test samples.

(2) Place each test sample of 35 ± 2 milligrams of energetic material in a loose pile in the center of the sandpaper.

b. Test other solid, granular materials by placing each test sample of 35 ± 2 milligrams of energetic material "as received" in a loose pile in the center of the sandpaper without further pretest processing.

40.1.2.3 Test procedure.

a. Place the test sample in the center of the anvil and lower the striker gently so that it rests on the top of the energetic pile.

b. Elevate the drop weight to a preselected height. Selection of the height used for the first drop is a matter of judgement. If the sensitivity of the test material has been previously measured, the first drop height will be chosen in the range where "fires" have occurred. If the material is of completely unknown sensitivity, an arbitrary starting height is used based on the sensitivity of similar compositions or the sensitivity which would be predicted from molecular structure.

c. Drop the weight and observe the result indicated by the noisemeter. If the neon lamp glows it is a "fire"; if not, the test is a "no-fire". The weight is caught by a sliding stop moved into position by the operator after initial rebound from contact with the striker. This prevents multiple impacts between weight and striker.

d. After the first fire is obtained (which may take 3 or 4 preliminary drops with an unknown material) successive drop heights are governed by the results of the previous drop according to the following procedure. Drop the weight from a height lower than the previous one by 0.093 log unit (where the log of a 10 centimeter drop is taken as 1.0). If the result is a fire, the next drop is 0.093 log unit lower; if a no-fire the next drop is 0.093 log unit higher. Continue testing by this "up and down" procedure for a total of 25 drops (usually called a "run").

e. After each drop, discard the test sample and use a fresh sample for the next drop. Clean the striker and anvil faces which are in contact with the test sample with solvent (such as acetone) after each test.

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f. Replace striker and anvil when working surfaces become roughened as determined by making carbon paper impressions of the surfaces. Old tools are refinished and reused. Replace the striker when its height has diminished by 0.250 inches.

40.1.3 Calculations.

a. For each test, record the log of the height from which the weight was dropped and the decision as to whether the drop resulted in a fire or no-fire.

b. Examine the data to determine whether more fires or no-fires occurred (if the numbers are equal, either may be used). Select the lower number for analysis and discard the balance of the data. Summarize statistically by the use of the following table (numbers are for illustration only):

log	i	n_i	$i \times n_i$	$i^2 \times n_i$
1.7	0	2	0	0
1.8	1	10	10	10
1.9	2	10	20	40

c. The log of a given drop height is entered in the first column. They are arranged in ascending order, starting with the lowest for which a test is recorded as indicated in the example above. In the next column, "i" is a consecutive number corresponding to the number of equal increments above the base, or "zero" line. The next column, "n_i", tabulates the number of fires (or no-fires) which occurred at i=0, i=1, i=2, etc. The other columns are computations of i times n_i and i^2 times n_i.

d. Compute the mean from the formula:

$$m = c + d\left(\frac{A}{N} \pm \frac{1}{2}\right)$$

Where:
$N = \sum n_i$
$A = \sum i \times n_i$
c = normalized height of the lowest line (i_0)
d = normalized interval between drops (0.093)

The sign inside the parenthesis is (+) if no-fires and (-) if fires are used.

e. Report the mean computed as the "50% point". It represents a 50% probability of fire.

f. The number may be reported in log units as determined directly from the computation. More often, the antilog is found and this is reported as a height in centimeters. Often, the standard deviation is also estimated by the following technique:

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A number, "M", is computed from the formula

$$M = \frac{\sum i^2 n_i}{N} - \left[\frac{i n_i}{N}\right]^2$$

Using a table or graph appearing in reference (2), a value "s" is obtained. The standard deviation (σ) is then:

$$\sigma = ds$$

It is always expressed in log units.

g. The table below sets out typical test results for eight common energetics.

	50% Point	
Energetic	cm	σ
Lead azide	4	0.12
PETN	12	0.13
RDX	24	0.11
HMX	26	0.10
Tetryl	38	0.07
Comp B	60	0.13
TNT	157	0.10
Explosive D	254	0.05

h. Reference for these calculations are as follows:

(1) Dixon, W.J. and Massey, F.J., Jr., "Introduction to Statistical Analysis", McGraw-Hill, 1951, pp 278-286.

(2) Applied Mathematics Panel of the National Defense Research Committee (AMP Report No. 101.1R SRG-P No. 40).

40.2 Friction pendulum.

40.2.1 Apparatus.

a. Pendulum friction device consists of:

(1) Steel swinging shoe (bob of the pendulum).

(2) Frame support of the pendulum (the "A-Frame").

(3) Two free bases of the "A-Frame" which can be raised or lowered simultaneously, controlling the pendulum swings. Test apparatus is shown in Figure 2.

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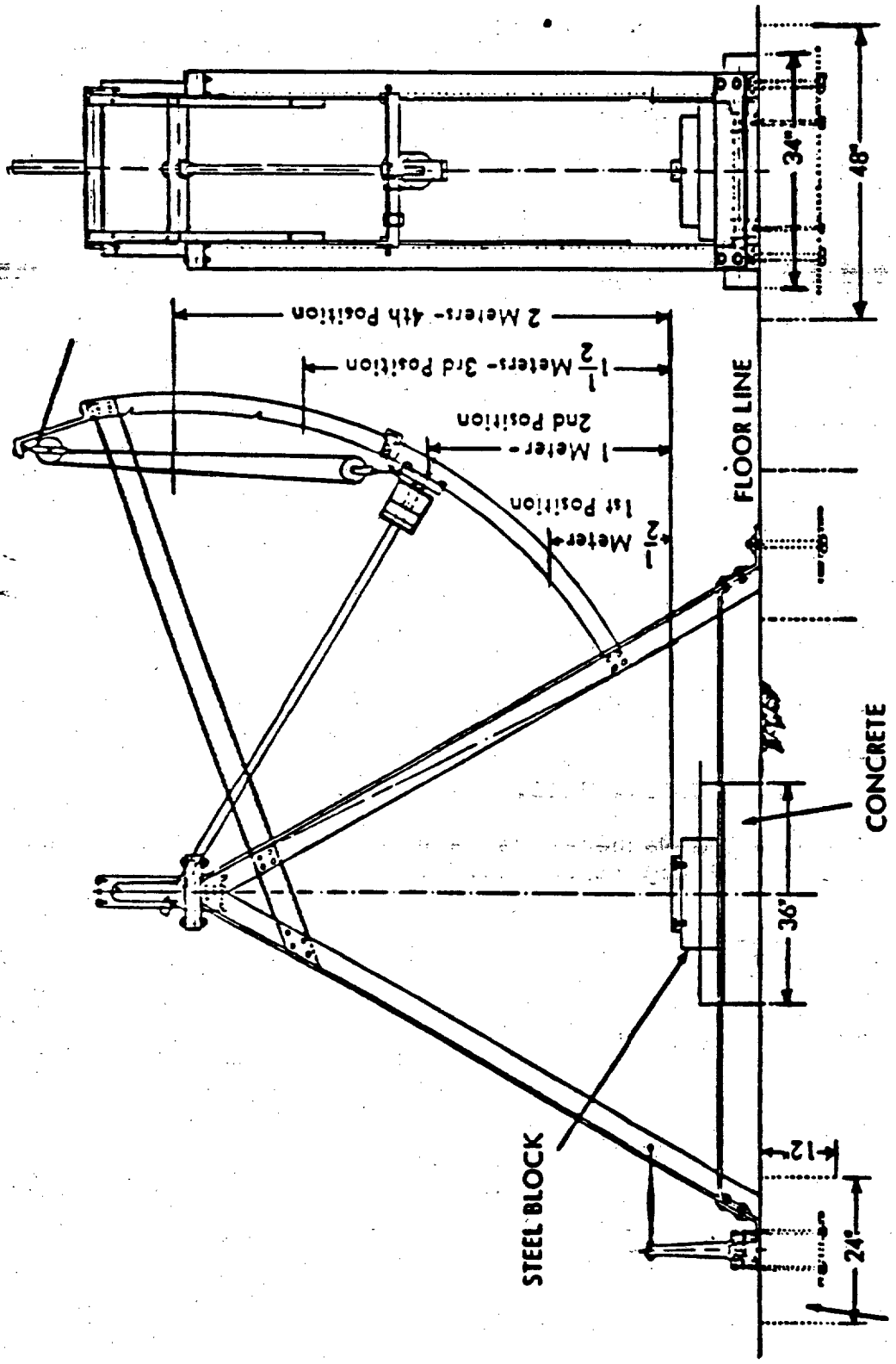


FIG. 2 PENDULUM FRICTION DEVICE ASSEMBLY

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b. The anvil consists of mild steel, Rockwell hardness B71 with a smooth, polished upper face. Three parallel grooves are cut across the central portion to prevent energetics from being brushed off.

c. Shoes, manufactured from mild steel, Rockwell hardness B71. Hard fiber for the fiber-faced shoe is procured in accordance with ASTM specification D-710-43T, commercial grade, density 1.34. It is supplied in three colors: black, red, and gray. It has been indicated that there may be slight differences in material for different colors. To make the test as uniform as possible, only the red is used. The fiber is not very resistant, and is therefore difficult to keep this shoe in first-class operating condition under the hard usage to which it is subjected. The condition of the shoe has to be inspected very often.

d. Automatic tripper, used to drop shoe from any desired vertical height ranging from 1/2 to 2 meters. Height of drop normally used at Picatinny Arsenal is 1 meter.

40.2.2 Procedure.

40.2.2.1 Preparation of apparatus.

a. Before trials are begun, test the shoe over the empty anvil to make sure that it is adjusted squarely with the anvil and that when loaded with weights it is adjusted in relation to the anvil so that it will swing across the face of the anvil 18 $\pm$ 1 times before coming to rest. Make the adjustment by operating the turnbuckle that raises or lowers the A-Frame.

b. The temperature of the anvil and shoe shall be controlled to be about 70 $\pm$ 5°F.

40.2.2.2 Test procedure.

a. With the device properly adjusted, 7 $\pm$ 0.1 grams of the energetic to be tested shall be spread in an even layer in and about the three grooves in the anvil and the shoe dropped upon it. After the pendulum has ceased swinging any remaining shall be brushed off, and the faces of the shoe anvil shall be cleaned thoroughly with a suitable solvent.

b. For protection during cleaning, the operator shall wear a face mask and asbestos gloves. The shoe and anvil shall be rubbed with carborundum cloth to remove any roughness caused by the impact of the shoe on the soft, steel anvil, also any adherent gritty matter. When the face of the shoe and anvil are clean and dry, another charge of 7 $\pm$ 0.1 grams of the energetic shall be placed on the anvil as before.

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c. The observations which have been made and recorded are "explosion" (whether complete or partial), "burning" (whether complete or partial), "local crackling" (whether distinctly audible or almost inaudible), and "no local crackling".

d. The number of swings on which these occurrences take place and the total number of swings for each trial shall be recorded. The number of swings on an empty anvil shall be checked and adjusted after every trial.

40.2.2.3 Evaluation and results.

a. A test shall consist of ten (10) trials with the steel shoe, except when complete explosion or burning occurs in any trial. If explosion or burning occurs, the trials with the steel shoe shall be discontinued.

b. Ten (10) trials shall be made with the fiber faced shoe only when complete explosion or burning occurs with the steel shoe, or as prescribed in the test directive. If the energetic passes the test with the use of the steel shoe, no further trials shall be conducted. From a viewpoint of economy and efficiency, therefore, all samples shall be tested first with the steel shoe.

c. An energetic material shall also be regarded as passing the friction pendulum test if in ten (10) trials with the fiber-faced shoe, there is no more than an almost inaudible local crackling, regardless of its behavior when subjected to the action of the steel shoe.

d. Typical test results.

Energetic	With fiber shoe	With steel shoe
PETN	Passed	Failed
RDX	Passed	Failed
Tetryl	Passed	Failed
Composition B	Passed	Failed
TNT	Passed	Failed
Explosive D	Passed	Failed

40.3 Electrostatic discharge (0.25 joule). This test shall be conducted in accordance with METHOD 4 of MIL-STD-1751.

40.4 Explosive temperature (5 seconds).

40.4.1 Apparatus. The apparatus for this test includes the following:

a. A barricaded press with a die for loading 40 milligrams of the energetic sample into a No. 8 blasting cap.

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b. A crimper.

c. A furnace with a molten woods metal bath and shield.

d. Associated temperature measurement and controls.

e. A time interval recorder.

40.4.2 Procedure.

40.4.2.1 Preparation of apparatus.

a. Check press operation and set loading force for 230 pounds with Dillion gauge.

b. Inspect equipment and empty blasting cap for cleanliness.

c. Place an empty blasting cap in the loading fixture.

d. Weigh out the proper amount of energetic material (40 mg), or use a previously calibrated conductive volumetric scoop, and pour into cap using a loading funnel.

e. Insert a copper gas check, open side up, into cap, and manually push the gas check down to bottom of cap with rod.

f. Put 30 mg of dried talc in cap, insert second gas check, and repeat step e.

g. Close barricade door and apply 6000 psi pressure (230 pounds loading force) for ten seconds.

h. Remove die from press. Behind a shield, remove cap from die with shielded tweezers and transfer cap to crimper.

i. Push cap into crimper with shielded tweezers and hold against stop.

j. Slowly lower handle of crimper until resistance is felt, then remove tweezers and crimp until stop is reached.

k. Raise handle of crimper and remove loaded cap with tweezers and place in nonpropagating tray behind barricade.

40.4.2.2 Test procedure.

a. Put a piece of masking tape around each cap.

b. Heat Woods metal bath to highest desired temperature. Stir bath with steel rod and wait at least five minutes until temperature is uniform.

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c. Rotate sample holder to loading position.

d. Reset timer to zero. Check start and stop signals, then reset zero.

e. Insert loaded cap in hole in fixture using shielded tweezers, and press into position. (Tape should hold cap firmly up, if not, add or remove tape as necessary).

f. Grasp handle and rotate fixture to allow sample to drop into bath.

g. When an explosion occurs, record time and temperature, and using handle, lift fixture and rotate back to the loading position.

h. If an explosion does not occur within two minutes, remove sample from the bath as described in step g and allow the sample to cool for at least one minute. Then destroy the sample; using shielded tweezers, place the cap behind a hood safety shield in a hole in a shielded steel block on a hot plate or in a furnace at 400°C to destroy the energetic material at the end of the test, allow sample to cool and then dispose of in an explosive waste can (underwater).

i. After all samples to be fired at the temperature of the bath have been fired, lower the temperature setting on the controller to the next temperature desired. Add cold metal, then stir thoroughly and check temperature. If still too hot, add more cold metal and stir again. Remove excess metal with a spoon and pour onto a steel plate to cool for future use. As long as the temperature is known at the time the sample is tested, the exact temperature set on the controller is not required.

j. Perform successive tests in the same manner to obtain time-to-explosion temperature values over a time range of 0.5 to 9 seconds.

40.4.2.3 Calculations.

a. Report the test results as the temperature in degrees centigrade which corresponds to the 5-second explosion temperature.

b. Determine the 5-second explosion temperature as follows. Plot the data on semilog paper, the logarithm of the time-to-explosion versus the reciprocal of the absolute temperature. This results in a straight line. A linear least square curve fit program is used to determine the temperature value for the 5 second time-to-explosion.

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40.5 Vacuum thermal compatability (stability 100°C).

40.6 Auto ignition temperature.

40.6.1 Apparatus.

a. Deltatherm II Differential Thermal Analysis Accessory or equivalent.

b. Mosely 7100B two-pen time-base recorder or equivalent.

40.6.2 Procedure.

a. Use a sample size which will produce an isosceles triangle of not too great an angle. Too steep an angle (80-90°) is not acceptable (Two large a sample); and too gradual an angle (0-35°) is not acceptable (too small a sample). Start with .2 to .5 milligrams for 10°/minute and 2 milligrams for 2.5°/minute.

b. Follow the general procedure for Differential Thermal Analysis, ASTM E474.

c. Place the energetic sample and the reference material into aluminum or steel cups (with size of up to 4.8 millimeters) and place into the thermo cups (steel cups to which an alumel and a chrome wire are welded).

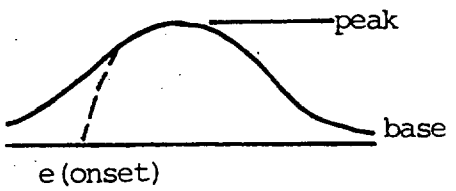
d. Close experimental area and set heating rate. The heating rates to be used shall be: 10, 5, 2 1/2, and 1 degrees centigrade per minute in each of four separate tests.

e. Set limits on recorder (paper speed, sensitivity) and instrument sensitivity. One pen shall record the temperature of the reference material; the other pen shall record the ΔT of the 2 cups at a space of 1/10 of an inch ahead of the temperature pen.

f. To minimuze the autocatalytic activity, pass a constant flow of argon across the sample holder while heating the sample at a constant rate.

40.6.3 Calculations.

a. Determine "e" onset and peak temperature by using the following guidelines.



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b. Graph ϕ/T^2 vs. T^{-1}, where ϕ = heating rate, $^{o}C/min$, T = peak temperature, to obtain the activation energy (slope).

c. The following relationship is used to solve for the necessary variables.

$$\frac{E\phi}{RT^2} = Ae^{-dE/RT} = 0.05 \text{ min.}^{-1}$$

d. Obtain A, the preexponential factor, at each peak temperature (at the four heating rates) and use the average of these values for the equation. This is done by solving for A in the above equation. (See attached excerpt from Thermochimica Acta, 14 (1976), 183-199).

e. Set $E\phi/RT^2$ equal to 0.05 and solve for the autoignition temperature from T in the exponent.

40.7 Differential thermal analysis (10% minimum). This test shall be conducted in accordance with ASTM-E474.

40.8 Rotary friction. This test shall be conducted in accordance with Method 506.1 of MIL-STD-1234.

50. PACKAGING. This section is not applicable to this appendix.

60. NOTES. This section is not applicable to this appendix.

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EXCERPT FROM THERMOCHIMICA ACTA

CALCULATION OF ACTIVATION ENERGY AND AUTOIGNITION TEMPERATURE FOR TNT (CONFINED) KNK 11-270

Temperature (°C)	Temperature (K)	Rate = ϕ (°C min^{-1})	$1/T_{(K)} + 10^3$	$\phi/T^2 \times 10^{-5}$
Temperature at the peak of the exotherm:				
333	606	22.0	1.650	5.847
315	588	10.7	1.701	3.058
296	569	5.3	1.757	1.637
287	560	2.6	1.786	0.829
Temperature at the beginning of the slope to the exotherm peak:				
326	599	22.0	1.666	5.988
309	582	10.7	1.718	3.121
292	565	5.3	1.770	1.660
285	558	2.6	1.792	0.835

$$\text{Slope} = \frac{E}{Ln\ R} = \frac{\log(\phi/T^2)_1 - (\phi/T^2)_2}{1/T_1 - 1/T_2}$$

E_p = 24200 cal mol^{-1}; E_s = 24900 cal mol^{-1}.
E_p = activation energy at peak; E_s = activation energy at slope;
ϕ = constant heating rate; T = temperature (K); R = gas constant;
A = frequency factor (sec^{-1}).

Peak temp. (K)	$\phi/T^2 \times 10^{-5}$	$\frac{E\phi}{RT^2}$	$e^{-E/RT} \times 10^{-9}$	$A \times 10^8$
606	5.991	0.72965	1.865	3.912
588	3.095	0.37695	1.013	3.721
568	1.642	0.1998	0.488	4.094
560	0.829	0.10096	0.358	(2.821)
			average (3) =	3.90×10^8
Slope temp. (K)				
599	6.131	0.76799	0.830	9.253
582	3.159	0.39571	0.451	8.774
565	1.660	0.20794	0.235	8.849
558	0.834	0.10447	0.178	(5.869)
			average (3) =	8.96×10^8

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$$\frac{E\phi}{RT^2} = Ae^{-E/RT}$$

as $\phi = 0$
$E\phi/RT^2 \longrightarrow 0.05$

$$\log 0.1 = \log 3.90 \times 10^8 - \frac{E}{RT \times 2.303}$$

T = 552K
279.0°C peak temperature

$$\log 0.05 = \log 8.96 \times 10^8 - \frac{E}{RT \times 2.303}$$

T = 547.6K
= 274.6°C autoignition temperature confined.

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